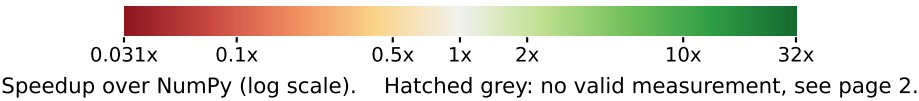


# Pluto vs DaCe auto\_optimize on the non-PolyBench NPBench kernels

Median speedup over NumPy (50 repetitions)

|                          | Pluto    | DaCe<br>auto_opt | NumPy    |                                           |
|--------------------------|----------|------------------|----------|-------------------------------------------|
| geo-mean                 | 1.8x     | 7.7x             | 315.4 ms |                                           |
| cholesky2                | 0.47x    | 1.7x             | 21.8 ms  | PolyBench family - no Pluto reference yet |
| covariance               | 2.9x     | 0.75x            | 28.0 ms  |                                           |
| covariance2              | 0.58x    | 0.14x            | 5.5 ms   |                                           |
| doitgen                  | 0.15x    | 46.5x            | 238.6 ms |                                           |
| gesummv                  | 26.8x    | 10.2x            | 197.3 ms |                                           |
| jacobi_1d                | 18.4x    | 3.2x             | 257.9 ms |                                           |
| k2mm                     | 0.09x    | 1.9x             | 167.8 ms |                                           |
| k3mm                     | 0.06x    | 1.1x             | 177.7 ms |                                           |
| symm                     | 7.1x     | 141x             | 7.27 s   |                                           |
| conv2d_bias              | declined | 38.1x            | 7.59 s   | deep learning                             |
| lenet                    | declined | 293x             | 1.46 s   |                                           |
| mlp                      | 0.02x    | 2.5x             | 23.7 ms  |                                           |
| resnet                   | 0.34x    | 68.0x            | 1.54 s   |                                           |
| softmax                  | 1.0x     | 11.6x            | 831.3 ms |                                           |
| hdiff                    | 6.4x     | 34.3x            | 152.6 ms | weather stencils                          |
| vadv                     | 11.4x    | 2.6x             | 544.4 ms |                                           |
| cavity_flow              | 1.7x     | 1.7x             | 1.61 s   | physics / CFD                             |
| channel_flow             | 2.9x     | 1.2x             | 2.86 s   |                                           |
| contour_integral         | 0.04x    | 2.3x             | 492.0 ms |                                           |
| nbody                    | 1.5x     | 3.7x             | 297.8 ms |                                           |
| scattering_self_energies | declined | 4.5x             | 1.06 s   |                                           |
| arc_distance             | 59.2x    | 49.9x            | 463.1 ms | signal / integration                      |
| azimint_hist             | declined | 3.9x             | 14.8 ms  |                                           |
| azimint_naive            | 5.3x     | 48.0x            | 491.3 ms |                                           |
| compute                  | 4.8x     | 4.8x             | 436.1 ms |                                           |
| stockham_fft             | declined | 0.72x            | 95.5 ms  |                                           |
| crc16                    | 168x     | 114x             | 1.41 s   | misc / non-affine                         |
| go_fast                  | 1.4x     | 1.3x             | 108.6 ms |                                           |
| mandelbrot1              | 11.6x    | 77.5x            | 877.8 ms |                                           |
| mandelbrot2              | declined | 0.75x            | 365.5 ms |                                           |
| spmv                     | declined | 1067x            | 237.8 ms |                                           |



Pluto vs DaCe auto\_optimize on the non-PolyBench NPBench kernels: detailed results

Preset paper, NumPy/Pluto from job 4523913; DaCe re-measured at b27aaed2f (job 4525319). Every blank cell on page 1 is accounted for below; a speedup is given only where the result validated.

| kernel                   | NumPy runtime | DaCe column: dace_cpu_autoopt / b27aaed2f |               |                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|---------------|-------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |               | Pluto                                     | DaCe auto_opt | Pluto: polycrcc --tile --parallel --codegen-context=1 (Clan frontend), clang -O3 -march=native<br>validated note / reason a cell is blank |                                                                                                                                                                                                                                                                                                                                                              |
| cholesky2                | 21.8 ms       | 0.47x                                     | 1.7x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| covariance               | 28.0 ms       | 2.9x                                      | 0.75x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| covariance2              | 5.5 ms        | 0.58x                                     | 0.14x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| doitgen                  | 238.6 ms      | 0.15x                                     | 46.5x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| gesummv                  | 197.3 ms      | 26.8x                                     | 10.2x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| jacobi_1d                | 257.9 ms      | 18.4x                                     | 3.2x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| k2mm                     | 167.8 ms      | 0.09x                                     | 1.9x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| k3mm                     | 177.7 ms      | 0.06x                                     | 1.1x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| symm                     | 7.27 s        | 7.1x                                      | 141x          | yes                                                                                                                                       | Pluto: correct and tiled, but polycrcc marked no loop parallel -- this is a SINGLE-THREADED result. The kernel is an inherently sequential recurrence and both polycrcc frontends agree there is no parallelism to find, so the speedup is not comparable to the 72-thread rows above.                                                                       |
| conv2d_bias              | 7.59 s        | declined                                  | 38.1x         | DaCe only                                                                                                                                 | Pluto: Pluto transformation: polycrcc did not terminate within its timeout.                                                                                                                                                                                                                                                                                  |
| lenet                    | 1.46 s        | declined                                  | 293x          | DaCe only                                                                                                                                 | Pluto: Pluto transformation: polycrcc did not terminate within its timeout.                                                                                                                                                                                                                                                                                  |
| mlp                      | 23.7 ms       | 0.02x                                     | 2.5x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| resnet                   | 1.54 s        | 0.34x                                     | 68.0x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| softmax                  | 831.3 ms      | 1.0x                                      | 11.6x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| hdiff                    | 152.6 ms      | 6.4x                                      | 34.3x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| vadv                     | 544.4 ms      | 11.4x                                     | 2.6x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| cavity_flow              | 1.61 s        | 1.7x                                      | 1.7x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| channel_flow             | 2.86 s        | 2.9x                                      | 1.2x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| contour_integral         | 492.0 ms      | 0.04x                                     | 2.3x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| nbody                    | 297.8 ms      | 1.5x                                      | 3.7x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| scattering_self_energies | 1.06 s        | declined                                  | 4.5x          | DaCe only                                                                                                                                 | Pluto: Two defects. Clan transposes adi's scop parameters: its domain for v[0][i] says -t+N>=0 and -i+TSTEPS-2>=0 where the loops are t=1..TSTEPS and i=1..N-1, so polycrcc guards on TSTEPS and can write past the arrays. Hoisting the scalar setup fixes that, but polycrcc then still emits the back-substitution before the loop producing its p and q. |
| arc_distance             | 463.1 ms      | 59.2x                                     | 49.9x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| azimint_hist             | 14.8 ms       | declined                                  | 3.9x          | DaCe only                                                                                                                                 | Pluto: Two defects. Clan transposes adi's scop parameters: its domain for v[0][i] says -t+N>=0 and -i+TSTEPS-2>=0 where the loops are t=1..TSTEPS and i=1..N-1, so polycrcc guards on TSTEPS and can write past the arrays. Hoisting the scalar setup fixes that, but polycrcc then still emits the back-substitution before the loop producing its p and q. |
| azimint_naive            | 491.3 ms      | 5.3x                                      | 48.0x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| compute                  | 436.1 ms      | 4.8x                                      | 4.8x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| stockham_fft             | 95.5 ms       | declined                                  | 0.72x         | DaCe only                                                                                                                                 | Pluto: Two defects. Clan transposes adi's scop parameters: its domain for v[0][i] says -t+N>=0 and -i+TSTEPS-2>=0 where the loops are t=1..TSTEPS and i=1..N-1, so polycrcc guards on TSTEPS and can write past the arrays. Hoisting the scalar setup fixes that, but polycrcc then still emits the back-substitution before the loop producing its p and q. |
| crc16                    | 1.41 s        | 168x                                      | 114x          | yes                                                                                                                                       | Pluto: correct and tiled, but polycrcc marked no loop parallel -- this is a SINGLE-THREADED result. The kernel is an inherently sequential recurrence and both polycrcc frontends agree there is no parallelism to find, so the speedup is not comparable to the 72-thread rows above.                                                                       |
| go_fast                  | 108.6 ms      | 1.4x                                      | 1.3x          | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| mandelbrot1              | 877.8 ms      | 11.6x                                     | 77.5x         | yes                                                                                                                                       | both columns validated against the NumPy reference                                                                                                                                                                                                                                                                                                           |
| mandelbrot2              | 365.5 ms      | declined                                  | 0.75x         | DaCe only                                                                                                                                 | Pluto: Two defects. Clan transposes adi's scop parameters: its domain for v[0][i] says -t+N>=0 and -i+TSTEPS-2>=0 where the loops are t=1..TSTEPS and i=1..N-1, so polycrcc guards on TSTEPS and can write past the arrays. Hoisting the scalar setup fixes that, but polycrcc then still emits the back-substitution before the loop producing its p and q. |
| spmv                     | 237.8 ms      | declined                                  | 1067x         | DaCe only                                                                                                                                 | Pluto: Two defects. Clan transposes adi's scop parameters: its domain for v[0][i] says -t+N>=0 and -i+TSTEPS-2>=0 where the loops are t=1..TSTEPS and i=1..N-1, so polycrcc guards on TSTEPS and can write past the arrays. Hoisting the scalar setup fixes that, but polycrcc then still emits the back-substitution before the loop producing its p and q. |